# ADVANCED CROWD MONITORING SYSTEM USING ARDUINO NANO

**20EC6203 - ELECTRONIC DESIGN PROJECT II**

**A PROJECT REPORT**

***Submitted by***

**SANJAY A**

***in partial fulfillment for the award of the degree***

***of***

**BACHELOR OF ENGINEERING**

***in***

**ELECTRONICS AND COMMUNICATION ENGINEERING**

**K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY**

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE, New Delhi)

**SAMAYAPURAM - 621 112**

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**(AUTONOMOUS)**

**SAMAYAPURAM, TIRUCHIRAPPALLI – 621 112**

## BONAFIDE CERTIFICATE

Certified that this project report titled **"ADVANCED CROWD MONITORING SYSTEM USING ARDUINO NANO"** is a Bonafide work of **SANJAY A (811722106091),** who carried out the project under my supervision. Certified further, that to the best of our knowledge the work reported here in does not from part of any other project report or dissertation the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

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## DECLARATION

We jointly declare that the project report on "**ADVANCED CROWD MONITORING SYSTEM USING ARDUINO NANO**" is the result of original work done by us and best of our knowledge, similar work has not been submitted to "**ANNA UNIVERSITY CHENNAI**" for the requirement of Degree of **BACHELOR OF ENGINEERING**. This project report is submitted on the partial fulfillment of the requirement of the award of Degree of **BACHELOR OF ENGINEERING.**

**Signature**

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Place: Samayapuram

Date:

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# ABSTRACT

In recent years, effective crowd management has become an essential aspect of public safety and event organization, especially in high-density areas such as transportation hubs, religious gatherings, stadiums, and public events. This project proposes an Advanced Crowd Monitoring System utilizing the Arduino Nano microcontroller to provide a cost-effective, real-time solution for detecting and managing crowd density. The system employs infrared (IR) sensors or ultrasonic sensors to count the number of individuals entering and exiting a monitored area. These sensors are strategically positioned at entry and exit points and are interfaced with the Arduino Nano, which acts as the central processing unit. By computing the difference between the number of people entering and exiting, the system continuously calculates the current crowd density in the area. To enhance functionality, the system integrates an LCD display or LED indicators to provide visual alerts when a predefined crowd threshold is exceeded. Optionally, a buzzer can be included to trigger remote alerts to authorities or organizers. The compact size and low power consumption of the Arduino Nano make the system easily deployable and scalable for various environments. Utilizing IR, the system accurately counts the number of people entering and leaving a designated area. The Arduino Nano processes this data to determine the current crowd level and displays it through an LCD indicators. When the crowd exceeds a predefined limit, the system can trigger visual or audible alerts. This enhances safety by enabling timely interventions during overcrowding situations.

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## LIST OF ABBREVIATIONS

| | | |
|---|---|---|
| **ACM** | - | Advanced Crowd Monitoring |
| **DC** | - | Direct Current |
| **GSM** | - | Global System for Mobile Communications |
| **IOT** | - | Internet of Things |
| **IR** | - | Infrared |
| **LCD** | - | Liquid Crystal Display |
| **LED** | - | Light Emitting Diode |
| **MCU** | - | Microcontroller Unit |
| **NANO** | - | Arduino Nano Board |
| **PIR** | - | Passive Infrared |
| **RX** | - | Receiver |
| **TX** | - | Transmitter |
| **UNO** | - | Arduino UNO Board |
| **USB** | - | Universal Serial Bus |

# CHAPTER 1

# INTRODUCTION

## 1.1 INTRODUCTION

In this fast-changing world of today, crowd management is one of the most important determinants for maintaining public safety, particularly in high-density populated areas like shopping centers, places of worship, transportation stations, and event arenas. Overcrowding can result in chaos, discomfiture, and even fatal conditions if left unattended. In order to tackle this problem, we suggest an Advanced Crowd Monitoring System based on Arduino nano that offers real-time crowd density detection and alerts through a buzzer instantly when threshold values are crossed.This project takes advantage of the Arduino nano capabilities, combined with sensors like ultrasonic or infrared (IR) sensors, to monitor continuously the number of individuals in a given area. When the crowd density exceeds a predetermined safe threshold, the system triggers a buzzer alarm to notify authorities or security personnel so that timely intervention and crowd control can be exercised. It is designed to detect and manage overcrowding in real-time. By using IR or ultrasonic sensors, the system counts the number of people entering and exiting a designated area. When the crowd exceeds a predefined limit, the Arduino triggers a buzzer to alert authorities instantly. This helps prevent dangerous situations caused by overpopulation and ensures public safety. The system is low-cost, efficient, and ideal for deployment in areas such as public transport stations, event halls, and religious places.Our solution is cost-effective, scalable, and deployable in both indoor and outdoor settings. It is designed with automation and safety in mind and is intended to help prevent overcrowding-related accidents by offering an early warning system, thus complementing public safety efforts.

## 1.2 BACKGROUND

Due to growing urbanization and population numbers, controlling massive crowds has also become a problem in public places. Overcrowding poses various safety risks such as stampedes, slow responses to emergencies, and overall distress for people present in the venue. Conventional crowd control procedures tend to focus on manual observation, which tends to be sluggish, ineffective, and susceptible to human mistakes. To address these issues, there is an increased demand for intelligent, automated systems that can detect real-time crowd densities and notify authorities when levels are reached.

The Arduino nano, an open-source microcontroller platform, provides an easy yet robust solution for creating such automation systems. If used together with sensors and alerting mechanisms such as buzzers, it can be programmed to count constantly the individuals passing into and out of a building and react the moment occupancy has surpassed safety limits. This project seeks to promote safety and situational awareness using a trusted, affordable, and scalable monitoring system adaptable to different environments.The conventional crowd control and monitoring processes are mostly dependent on human observation, CCTV cameras, and manual headcounts, which are time-consuming, resource-intensive, and tend to be inaccurate. Furthermore, such processes do not provide instant alerts or automatic intervention as soon as a situation reaches unsafe levels.

The Arduino nano is a popular, open-source ATmega328P-based microcontroller board that presents a great vehicle to construct these kinds of systems. It's user-friendly, economical, and easily customizable and therefore suitable for learning and for-practical purposes. When used with sensors like Infrared (IR) or Ultrasonic sensors, Arduino can be made to track the number of individuals entering or leaving a given area. With a running total of occupancy, the system can ascertain whether the number of people is more than a pre-programmed safe capacity.

# CHAPTER 2

# LITERATURE SURVEY

Anita Sharma et al., 2024 explored a crowd monitoring system utilizing Arduino Nano integrated with ultrasonic sensors and IoT for real-time people counting and congestion detection. Their research emphasizes the benefits of low-cost microcontroller-based solutions in densely populated areas for effective crowd control.

Sharma et al., 2024 developed a crowd monitoring system utilizing Arduino Nano integrated with ultrasonic sensors and IoT for real-time people counting and congestion detection. Their research emphasizes the benefits of low-cost microcontroller-based solutions in densely populated areas for effective crowd control. However, the study also notes limitations in sensor accuracy under variable lighting and environmental conditions.

Krisna et al., 2023 implemented an IoT-based crowd monitoring device using Arduino UNO R4 and IoT Cloud. The system employs IR sensors and servo motors at entry and exit points to monitor crowd density. While the approach offers real-time monitoring capabilities, the study identifies challenges in sensor calibration and the need for robust data security measures. The author suggests enhancing sensor accuracy through calibration algorithms and implementing encryption protocols to safeguard transmitted data.

Priya Vihari et al., 2023 developed a bidirectional visitor counter using Arduino and ultrasonic sensors to monitor foot traffic in retail stores. The system provides real-time data visualization on an LCD display. However, the study notes challenges in sensor calibration and the need for scalability in larger installations. The author recommends implementing machine learning algorithms for adaptive calibration and exploring wireless communication options for scalability.

Fahad et al., 2022 designed a bidirectional visitor counter using Arduino and ultrasonic sensors to track the number of people entering and exiting a specific area. The system utilizes an LCD display to show real-time counts. Despite its simplicity, the study highlights issues related to sensor placement and environmental factors affecting accuracy. The author proposes exploring alternative sensor technologies and optimizing placement strategies to improve reliability.

Varun Nair et al., 2022 implemented a visitor counting system using Arduino and ultrasonic sensors to monitor crowd density in public spaces. The system employs an OLED display to show real-time counts. While the approach offers cost-effective monitoring, the study identifies limitations in sensor accuracy under varying environmental conditions. The author suggests integrating additional sensor modalities and implementing data fusion techniques to enhance accuracy.

Ahmed et al., 2020 developed an IoT-based real-time noise mapping system using Arduino Nano and ESP8266 modules. The system measures ambient sound intensity and visualizes data on a web application. Despite its potential, the study identifies challenges in sensor calibration and data accuracy.

Karar et al., 2020 introduced GASDUINO, a wireless air quality monitoring system using Arduino microcontrollers and gas sensors. The system provides real-time air quality data visualization through an Android application. While the approach offers valuable environmental monitoring, the study notes challenges in sensor calibration and data accuracy.

Mumtaz et al., 2018 proposed an automation system for controlling streetlights and monitoring objects using Arduino. The system employs sensors to detect objects and control streetlight operations accordingly. While the approach offers energy-efficient solutions, the study identifies challenges in sensor accuracy and system scalability. The authors suggest exploring alternative

sensor technologies and implementing scalable architectures for broader applications..

Ritika Sinha et al., 2018 design an Arduino-based people counting system equipped with a buzzer that activates when the room reaches full capacity. The project is aimed at improving safety in malls and conference halls. Their findings demonstrate that buzzer alerts are effective in notifying security staff quickly and can be deployed easily without advanced technical infrastructure.

The literature reviews above center around the development, implementation, and improvement of automated crowd monitoring systems using microcontrollers, particularly Arduino UNO, along with various sensors and alert mechanisms. These studies collectively aim to tackle the issue of overcrowding in public and private spaces, emphasizing real-time detection, and immediate alerting for safety and management purposes. the use of Arduino UNO and various sensors, such as infrared (IR) and ultrasonic sensors, for real-time occupancy detection. Most studies highlight the cost-effectiveness and simplicity of Arduino-based solutions for automated crowd management in public spaces. While several systems integrate IoT technologies for remote monitoring, many also focus on threshold-based crowd detection and immediate alert mechanisms.this system feature Notably, buzzer alerts have been recognized for their effectiveness in providing instant feedback, ensuring rapid responses to overcrowding situations. These studies underscore the importance of low-cost, accessible, and reliable solutions for public safety, particularly in environments where advanced infrastructure may be lacking.It is designed to detect and manage overcrowding in real- time. By using IR or ultrasonic sensors, the system counts the number of people entering and exiting a designated area. When the crowd exceeds a predefined limit, the Arduino triggers a buzzer to alert authorities instantly. This helps prevent dangerous situations caused by overpopulation and ensures public safety.

# CHAPTER 3

# EXISTING AND PROPOSED SYSTEM

## 3.1 EXISTING METHOD

Crowd monitoring is a critical aspect of managing large public gatherings, ensuring the safety and security of individuals in places such as stadiums, concert halls, shopping malls, or public events. The goal is to prevent overcrowding, stampedes, or any situation that may pose a threat to public safety. Over the years, several systems have been developed to monitor crowds, with varying degrees of efficiency and complexity. These systems generally fall into two broad categories: manual crowd monitoring and automated sensor-based systems. Both systems, while functional, have certain limitations.

Manual crowd monitoring traditionally relies on human operators to observe crowds and assess crowd density based on visual observation, CCTV footage, or physical patrols. Security personnel or event managers are tasked with tracking the number of people in various sections of the venue and ensuring that areas do not become overcrowded. In some cases, security personnel may also use walkie-talkies or other communication tools to relay crowd density information.

One major limitation of manual crowd monitoring is that it is highly labor-intensive and often unreliable. Relying on human judgment and observation can lead to delays in identifying crowd density issues, especially in large crowds or during peak times. Additionally, the human eye cannot consistently detect minor variations in crowd density that might indicate the potential for overcrowding. As a result, manual systems are reactive in nature, and interventions may only occur after dangerous situations, such as stampedes or suffocation, have begun to develop.

Furthermore, human-operated systems require a significant number of personnel to monitor large crowds, increasing operational costs. The need for continuous human presence also means that manual systems are not scalable,

particularly in large venues or events with thousands of attendees. Human error, fatigue, and miscommunication are other factors that contribute to the inefficiency of manual crowd monitoring.

As technology has advanced, many venues have turned to CCTV cameras for crowd monitoring. These systems allow security personnel to observe crowds remotely and from multiple angles. CCTV systems often feature facial recognition and motion tracking software, which can be used to detect crowd density in real time. Some systems can even detect unusual crowd movements or behaviors that may suggest a potential emergency or disorderly conduct.

Despite their advantages, CCTV surveillance-based systems also have significant drawbacks. First, while cameras can capture high-quality video footage, they are only effective if there is a sufficient number of cameras installed across the venue. Large venues may require hundreds of cameras to monitor every corner, leading to high installation and maintenance costs. Additionally, these systems rely heavily on human operators to interpret the video footage, meaning that real-time responses to crowd density issues may still be slow and dependent on the skill.

Another limitation of CCTV-based systems is that they are not designed to provide specific crowd density metrics. While cameras may detect large crowds in certain areas, they do not give an accurate measure of crowd size or the potential for dangerous congestion.

One of the primary benefits of sensor-based systems is that they can be integrated with software that provides real-time analytics, allowing for automatic monitoring of crowd conditions. These systems can trigger alerts if crowd density exceeds a predefined threshold, signaling potential risks such as overcrowding or the need for crowd control measures. However, these systems still require regular calibration and maintenance to ensure accuracy, especially in environments where conditions change frequently.automated response mechanisms, and real-time data analysis, limiting their overall effectiveness.

## 3.2 PROPOSED METHOD

The proposed method for the Advanced Crowd Monitoring System using Arduino nano is designed to provide an efficient and cost-effective solution for real-time crowd management in public places. This system aims to prevent overcrowding by constantly monitoring the number of individuals entering and exiting a designated area and generating an alert when a specific crowd threshold is exceeded. At the core of the system is the Arduino nano, a widely used and versatile development board known for its simplicity and ease of integration with various sensors. The primary components of the system include the Arduino nano, infrared (IR) or ultrasonic sensors for detecting movement, a buzzer for audio alerts, and optional components such as LEDs for visual alerts and an LCD display or serial monitor for displaying the current crowd count. The system operates by deploying two sensors at entry and exit points of a monitored area. These sensors work together to detect the direction of movement of individuals. When a person crosses the sensor setup, the system determines whether they are entering or exiting based on the sequence in which the sensors are triggered. If Sensor A (placed on the inside) is triggered before Sensor B (on the outside), the person is exiting the area, and the counter is decremented. Conversely, if Sensor B is triggered before Sensor A, the person is entering the area, and the counter is incremented. This logic helps in maintaining an accurate real-time count of the number of people within the area at any given time.

A crucial feature of the system is the predefined threshold that determines the maximum number of individuals allowed in the monitored space. This threshold can be set depending on the size and safety regulations of the venue. When the number of people within the area reaches or exceeds this limit, the system activates the alert mechanism. The buzzer emits a loud sound to notify concerned authorities or security personnel about the potential overcrowding. In addition to the buzzer, an LED can be used to provide a visual signal — for instance, turning

red when the threshold is breached. Simultaneously, a message can be displayed on an LCD screen, such as “Crowd Limit Exceeded,” to inform those nearby. The buzzer remains on as long as the crowd count remains above the limit and automatically turns off once the number drops below the set threshold. This approach allows for timely intervention and management of the situation, potentially avoiding hazards such as stampedes or other overcrowding-related incidents.

The entire system is powered through a 9V battery or USB connection, making it portable and easy to deploy in different environments. Furthermore, this system is scalable, meaning additional sensor sets can be introduced for wider areas or multiple entry and exit points, with slight modifications to the software logic. One of the key advantages of using Arduino nano is its compatibility with various communication modules, which makes it possible to extend the system in the future. For example, integration with GSM modules can allow the system to send SMS alerts to security personnel, or it can be linked to cloud-based IoT platforms for remote monitoring and analytics. Overall, this method provides a practical, real-time solution for managing crowd sizes effectively without the need for expensive or complex surveillance systems. It is particularly beneficial in areas with limited resources but a high need for crowd control, such as schools, small event halls, or religious gathering spots. By leveraging the simplicity and reliability of Arduino and basic sensors, the proposed method ensures that public safety and comfort are prioritized through continuous monitoring and timely alerts In addition to its core functionality, the system emphasizes ease of installation and adaptability, making it a versatile solution for a wide range of environments. Since the components used—such as IR sensors, buzzers, and Arduino nano—are readily available and affordable, the system is ideal for both small-scale applications and larger public facilities that may need budget-friendly crowd management solutions. The sensors are non-invasive, meaning they do not require physical contact with individuals, which maintains privacy and hygiene—an especially important consideration in the

post-pandemic era. Moreover, the Arduino nano allows for easy modifications and code updates, which makes the system highly customizable depending on the specific requirements of the venue, such as adjusting the maximum capacity limit or changing the type of alert notification.

### 3.2.1 BLOCK DIAGRAM

This project aims to design a real-time crowd monitoring system that detects, controls, and manages crowd density in a given area using Arduino nano. It uses RFID scanning for identification, IR sensors for physical movement detection, a predictive algorithm for analyzing crowd size, and buzzer alerts to prevent overcrowding situations. The entire process is modular, following the flow between blocks, each serving a specific purpose..

The Advanced Crowd Monitoring System is powered by a Li-ion battery, providing a stable and portable power supply to all the components. The heart of the system is the Arduino nano, which acts as the main controller, processing data from connected modules and managing system operations. Individuals are first scanned through an RFID scanner, where their entry or exit is logged. To verify the actual presence and movement of individuals, additional sensors (such as infrared or ultrasonic) detect physical motion, ensuring accuracy and preventing false entries. Upon successful verification, a servo motor controls the physical entry/exit gate, allowing or denying access based on the inputs received. The gathered data is then processed using a predictive algorithm which continuously estimates the crowd density inside the monitored area and predicts any risk of overcrowding. If the crowd size approaches unsafe levels, the system triggers an alert message to warn the authorities or users. This alert is also shown on a display unit, which continuously provides real-time updates about the number of individuals inside, warning messages like "Crowd Limit Reached," and instructions for users. This integrated system ensures efficient crowd management with predictive safety measures and instant alerts.

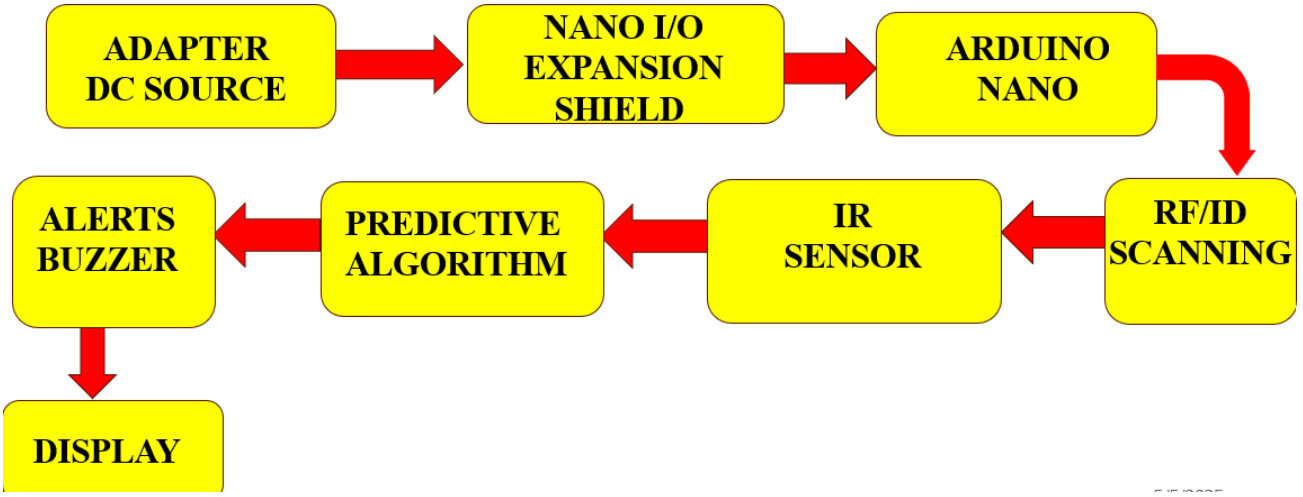



Figure 3.1 Block Diagram

The power adapter 12v serves as the primary power source for the entire system. It provides a stable and portable energy supply to ensure continuous operation without the need for external power connections. This makes the system highly mobile and reliable, especially in outdoor or large venue applications where wired connections are impractical. The battery powers not just the Arduino board but also all peripheral components like sensors, RFID modules, and display units.

The Arduino nano acts as the brain of the system. It processes all incoming data from RFID scanners and sensors, makes decisions based on programmed algorithms, and controls the servo motor and alert system accordingly. Additionally, the built-in module allows for wireless communication, making it possible to send alerts or updates remotely if needed. It is responsible for running the predictive algorithm to estimate crowd density and trigger necessary actions when thresholds are reached.

The RFID scanning block is responsible for identifying and logging the entry or exit of individuals. When a person approaches the system with an RFID card or tag, the reader scans the information and sends it to the Arduino for verification and recording. This ensures that every individual is accounted for properly, maintaining an organized and automated attendance or entry system. It

also helps link each person uniquely to the crowd monitoring data.

The sensors are used to detect the actual physical movement of individuals after RFID scanning. They confirm whether a person has indeed entered or exited the premises, reducing errors that may occur if someone scans their RFID card but doesn't physically pass through. Sensors like IR or ultrasonic types detect motion, providing a double-check mechanism that improves the accuracy and reliability of the crowd monitoring system.

The servo motor is used to control a barrier or gate mechanism based on the results from RFID scanning and sensor detection. Once a person is authenticated and verified, the servo motor automatically opens to allow entry or exit. It then closes immediately after, preventing unauthorized access. This automation ensures smooth crowd movement while maintaining security and access control within the monitored area.

The predictive algorithm continuously processes the data collected from the RFID scanner and sensors to estimate the number of people present inside the monitored zone. It forecasts potential risks by analyzing patterns like sudden increases in the crowd density. If the number of individuals exceeds a pre-set safe limit, the algorithm immediately flags the situation as critical, triggering alerts to warn authorities or system operators.

The alerts message block is activated when the predictive algorithm detects overcrowding or unusual activity. It generates warning notifications, which can be visual (e.g., flashing LEDs), audio (buzzers), or digital (sending messages via WiFi). These alerts ensure that necessary action can be taken immediately to prevent accidents or manage the crowd better. The message system enhances the safety and responsiveness of the entire monitoring setup.

The display unit continuously provides real-time information about the crowd status. It shows the current count of individuals inside, any warning or alert messages, and instructions for users (like "Please Wait" or "Entry Allowed"). The display ensures that both users and operators are well-informed at all times,

facilitating smooth operations and improving situational awareness in critical crowd management scenarios.

### 3.2.2 CIRCUIT DIAGRAM

The Advanced Crowd Monitoring System is designed to efficiently manage and control crowd density in sensitive areas. At the heart of the system, the Arduino nano microcontroller coordinates inputs from sensors and modules to track the number of people accurately. IR sensors are installed at entry and exit points to detect human movement, allowing the Arduino to increment or decrement the crowd count based on the direction of movement. An RFID module is used to scan and verify authorized individuals before allowing entry, adding a layer of security to the system. The LCD display plays a crucial role in providing real-time updates such as the current crowd count, access verification messages, and warning alerts.

To enhance safety, the system is equipped with a buzzer and LED indicator that get activated if the crowd exceeds the predefined limit. The buzzer provides an audible warning while the LED offers a visual alert, ensuring immediate attention to overcrowding situations. This dual alert mechanism ensures that both sound and sight warnings are available for maximum effectiveness. Additionally, a wireless module like ESP8266 or NRF24L01 can be integrated to send real-time data to a remote monitoring unit, allowing authorities to oversee crowd conditions from a distance.

Overall, this project offers a smart, automated, and reliable method for crowd control. It reduces human dependency, minimizes the chances of overcrowding-related incidents, and provides a structured flow of people through secure scanning and monitoring. The system is ideal for use in public spaces such as malls, airports, hospitals, and event venues. In operation, the system works sequentially: First, the RFID tag is scanned and verified. If accepted, the IR sensor

detects the movement and updates the crowd count. The LCD displays the updated information. If the crowd exceeds the limit, both the buzzer and LED are triggered to issue an alert. Meanwhile, if wireless transmission is implemented, data is sent to a central monitoring system for real-time tracking.

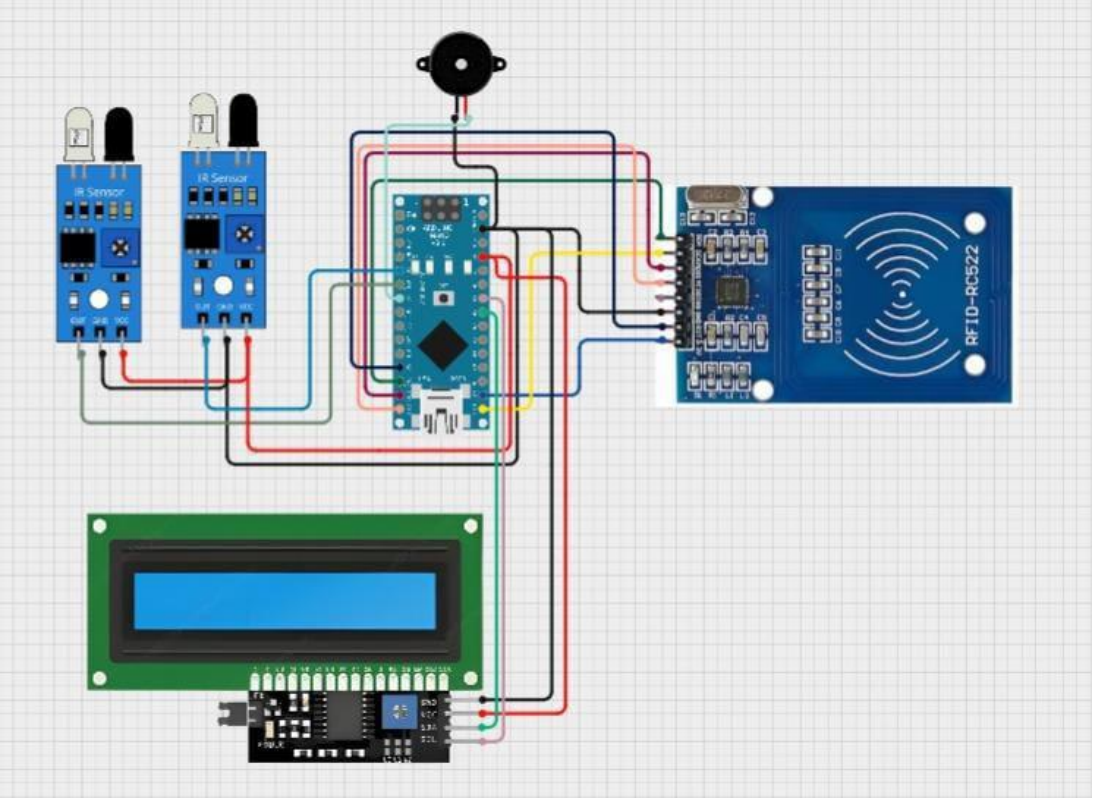
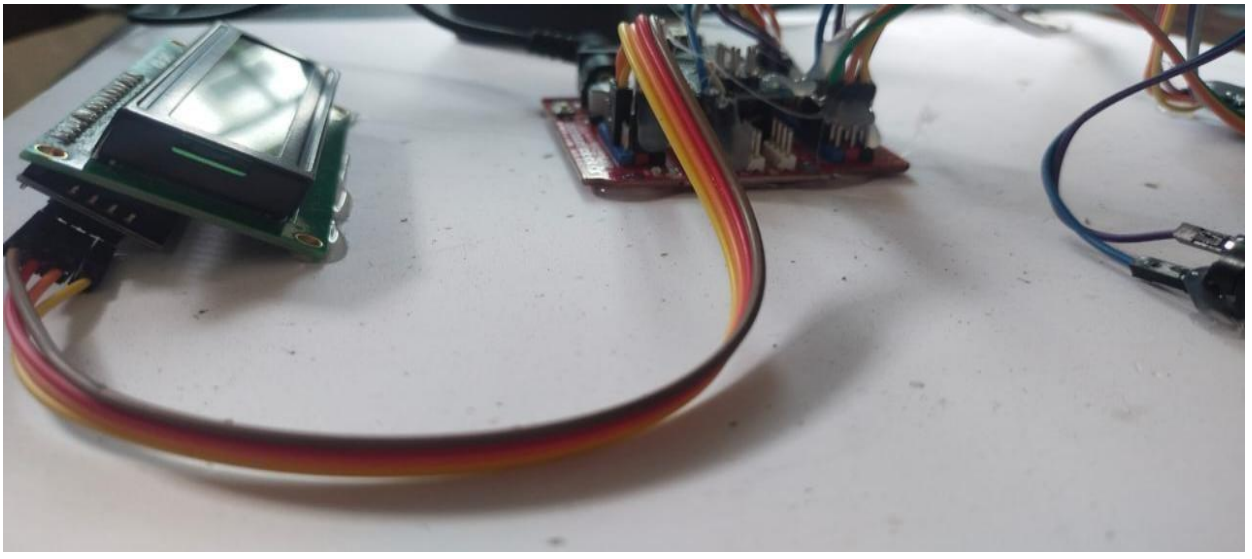



Figure 3.2 Circuit Diagram

### 3.2.3 WORKING MODEL



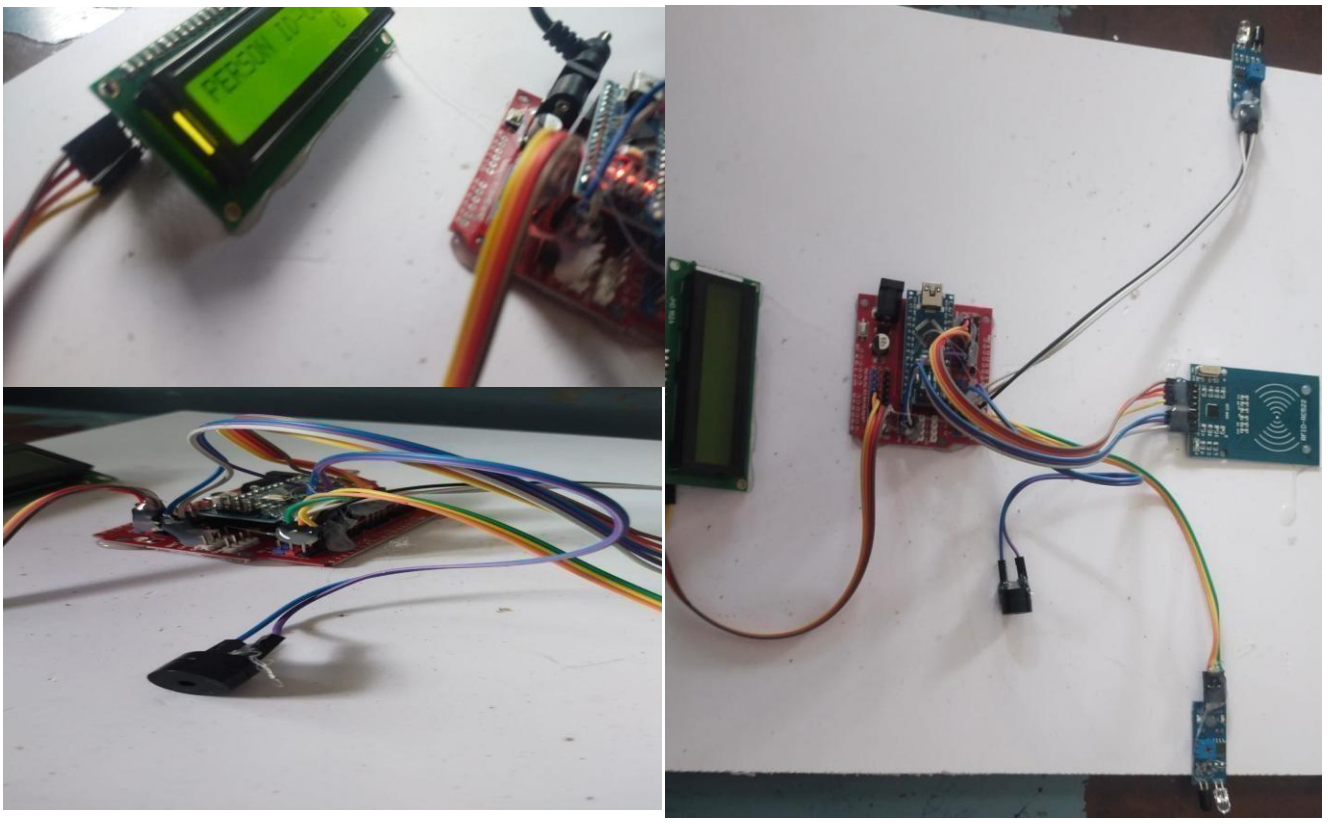


Figure 3.3 Working Model

### 3.2.4 APPLICATIONS

1. **Temple Management –** Monitors the number of devotees entering and exiting to avoid overcrowding during festivals. Triggers buzzer alerts when safe limits are exceeded, helping maintain discipline and prevent stampedes. It is mainly used for this management because of monitoring the crowd in temples.

2. **Shopping Malls –** Tracks foot traffic at entrances using IR sensors. Automatically sounds an alert when crowd levels cross the threshold, ensuring safety compliance and enhancing customer experience. It enhances the security purpose for the shopping malls and theatres.

3. **School Assembly Halls –** Keeps count of students entering closed areas like auditoriums or exam halls. Prevents over-occupancy and maintains fire safety standards through buzzer-based alerts.

4. **Public Transport Terminals –** Deployed at metro stations or bus stops to monitor passenger flow. When crowd size becomes unsafe, the buzzer activates to signal staff for crowd dispersal or entry restriction.

5. **Hospitals –** Installed in outpatient waiting areas to manage patient flow. Alerts when seating or room capacity is full, improving health safety and reducing the risk of infection spread.
6. **Event Venues** – Used at the entrance of concerts, exhibitions, or conferences. Helps event staff monitor live attendance and triggers alerts once capacity is reached, preventing gate rush or security issues.

7. **Corporate Offices –** Manages occupancy in meeting rooms or office spaces. Sends alerts when room limits are exceeded, aiding in social distancing protocols or energy-saving automation.

8. **Airports --** Monitors passenger density in check-in, security, and boarding areas using video surveillance and AI. Sends real-time alerts to staff for queue management and congestion control.

9. **Manufacturing Plants --** Uses RFID tags to track employee movement in real time. Triggers alerts when restricted zones are overcrowded, improving worker safety and operational efficiency.

10. **Hospitals --** Monitors patient and visitor movement in waiting areas using IR and thermal sensors. Helps control overcrowding and maintain social distancing during health emergencies.

11. **Schools & universities --** Tracks student entry and exit through RFID ID cards. Prevents overcrowding in corridors and classrooms, improving campus safety and attendance accuracy.

### 3.2.5 ADVANTAGES

1. **Real-Time Crowd Control –** Continuously tracks the number of people in a specific area using IR sensors. Triggers a buzzer alert when the preset capacity is exceeded, allowing quick action to prevent overcrowding.

2. **Low-Cost Solution –** Utilizes affordable components like Arduino Uno and basic sensors. Ideal for schools, small public venues, or budget-restricted projects that need effective crowd monitoring.

3. **Accurate People Counting –** Uses directional sensing logic between entry and exit. Minimizes errors in counting and ensures precise occupancy data.

4. **Automated Alert System –** Activates an audible buzzer when maximum capacity is reached. Eliminates the need for manual supervision and improves response time to overcrowding.

5. **Easy Integration –** Compatible with additional modules such as GSM, Wi-Fi, or LCD displays. Can be expanded to send SMS alerts or connect with mobile apps for remote monitoring.

6. **Energy-Efficient Operation –** Consumes minimal power and can run on batteries or USB. Suitable for continuous usage in low-resource or off-grid locations.

7. **Portable and Flexible –** Compact design allows deployment in temporary venues like events or fairs. Can be moved and reconfigured easily as needed.

8. **Enhanced User Experience** -- Reduces waiting time, manages queues better, and improves overall flow in public spaces like malls, airports, or events.

# CHAPTER 4

# SOFTWARE AND HARDWARE REQUIREMENTS

## 4.1 HARDWARE REQUIREMENTS

### 4.1.1 ARDUINO NANO

The Arduino Nano is a compact and powerful microcontroller board developed for embedded systems and DIY electronics projects. It is based on the ATmega328P microcontroller, the same chip used in the widely known Arduino Uno, but it comes in a much smaller and breadboard-friendly form factor. the Arduino Nano offers robust functionality in a miniature size. It operates at a voltage of 5V and has a clock speed of 16 MHz, allowing it to execute instructions quickly and efficiently. The board features 14 digital I/O pins, 6 of which support PWM (Pulse Width Modulation), and 8 analog input pins, making it versatile for interfacing with sensors, actuators, and other electronic components.

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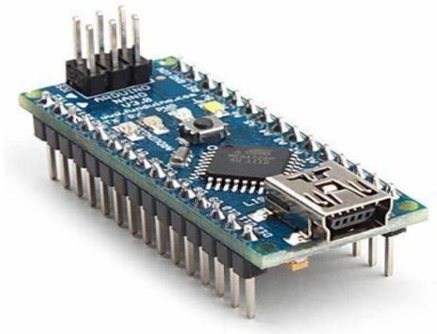

Figure 4.1 Arduino Nano

The board can be powered via Mini-USB cable or through the VIN and GND pins using an external power source (7-12V recommended). It communicates with the computer or other devices using a USB-to-serial converter, typically a CH340 or FT232 chip depending on the manufacturer. Programming the Nano is simple using the Arduino IDE, and it supports C language with a large number of readily available libraries and community resources.

### 4.1.2 NANO IO EXPANSION SHIELD

The Arduino Uno is a widely used microcontroller board based on the ATmega328P. It serves as the central processing unit in electronic projects, enabling users to interface with sensors, actuators, and other electronic components. Featuring multiple digital and analog input/output pins, the Arduino Uno supports a variety of applications, from simple LED control to complex automation systems. Its compatibility with the Arduino IDE simplifies programming and debugging. The board operates at 5V and features 14 digital input/output pins (six of which can be used as PWM outputs), six analog input pins, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header, and a reset button. It can be powered via a USB connection or an external power supply ranging from 7V to 12V. The onboard voltage regulator ensures a stable 5V output for connected components.

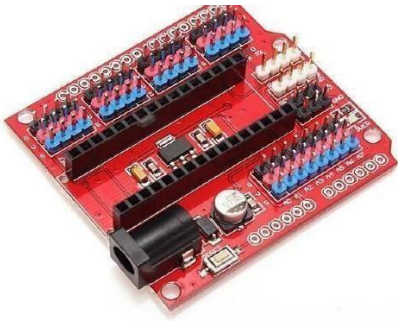

Figure 4.2 Nano Io Expansion Shield

One of the key features of the Nano I/O Expansion Shield is the inclusion of additional connectors and interfaces. Most shields offer 3-pin headers (VCC, GND) for each I/O pin, making it ideal for directly connecting servos and sensors without additional wiring. Some versions also include I2C and UART interfaces, power terminals, reset buttons, and external power input options, adding further versatility to the Nano. This layout helps reduce wiring errors and simplifies circuit building, when prototyping more complex projects.

### 4.1.3 IR SENSOR

An infrared (IR) sensor is a component that responds to infrared radiation from objects around it. Infrared radiation is a form of electromagnetic wave that lies slightly beyond the spectrum of visible light and is generally thought of as heat energy. IR sensors are utilized in a broad range of applications for object sensing and detection, measuring temperature, and facilitating remote communication. Infrared sensors function in two broad categories, namely active and passive. Active IR sensors send out their own infrared light and measure its reflection from objects close to them to determine distance or detect the presence of an obstacle. Passive IR sensors (illustrated in Figure 4.1.3), however, do not send out any radiation but rather receive the infrared energy that objects naturally emit, thus being suitable for use in motion detection.

Figure 4.3 Infrared Sensor

These sensors find applications across a wide range of fields. In consumer electronics, they are used in TV and air conditioner remote controls, among other devices. In security systems, passive IR sensors play a vital role in alarm and surveillance motion detection. IR sensors are used for quality monitoring, and temperature sensing in industrial automation systems. R sensors operate at 3.3V or 5V, making them compatible with Arduino boards. They often have three pins: VCC, GND, and OUT (signal). In addition, IR sensors are also utilized in robotics for object detection and navigation, and in the field of medicine for thermal imaging and non-contact temperature measurements.

### 4.1.4 PIEZO BUZZER

A piezo buzzer is a compact, low-cost, and efficient audio signaling device commonly used in electronic systems to produce sound or alarms. It works based on the piezoelectric effect, where certain materials (like piezoelectric ceramics) generate mechanical vibrations (sound) when an electric voltage is applied to them. These vibrations create audible tones or beeps, making piezo buzzers ideal for use in devices that require simple sound output**.** Piezo buzzers come in two types: active and passive. An active piezo buzzer has a built-in oscillator and produces sound when connected to a power supply, usually 5V, without requiring any control signal. A passive piezo buzzer, on the other hand, needs an external frequency signal (such as a PWM signal from a microcontroller) to produce sound, making it suitable for generating custom tones and melodies. When the sensor data indicates that the number of people exceeds a pre-set safe limit, the microcontroller (such as Arduino Nano, ESP32, or Raspberry Pi) triggers the buzzer to sound an alert.

Figure 4.4 Piezo Buzzer

This direct sound output is especially useful in environments like public venues, transport terminals, stadiums, shopping malls, or religious gatherings, where visual indicators alone may not be noticed promptly. it improves response time, enhances safety, and helps prevent hazardous overcrowding or panic situations. The advantages of ultrasonic sensors are significant. They are capable of functioning in low-light environments.

### 4.1.5 LCD DISPLAY

An LCD display is widely used in Arduino projects to provide a user interface for displaying information such as sensor data, system status, or interactive messages. The most commonly used type is the 16x2 character LCD, which can show two lines of 16 characters each. It operates using liquid crystals that control light transmission to render text and symbols. The LCD can be interfaced with Arduino using a parallel connection or an I2C module, with the latter reducing the number of required pins. In a standard setup, the LCD connects via GPIO pins for data transmission, enabling the Arduino to send commands and characters for display. The integration of an LCD enhances project visibility, making it useful in applications such as temperature monitoring, automation interfaces, and IoT systems. By combining Arduino's processing capabilities with LCD output, projects become more interactive and user-friendly, improving real-time feedback and usability.

Figure 4.5 Lcd Display

The display can be connected to the Arduino either through a parallel interface, requiring multiple GPIO pins, or an I2C module, which simplifies wiring and reduces pin usage. Arduino communicates with the LCD by sending commands and data, enabling clear visual representation of measurements, text, or even simple animations. This makes it a valuable tool in projects ranging from home automation and IoT devices to robotics and industrial monitoring systems. Whether used for creating interactive user interfaces

### 4.1.6 RF ID SCANNING

RFID scanning technology has revolutionized crowd monitoring by enabling real-time tracking and efficient management of large gatherings. Through RFID tags embedded in smart cards or wearable devices, individuals can be monitored seamlessly as they move through designated areas. Advanced scanners placed at strategic locations collect data, allowing authorities to analyze movement patterns, detect overcrowding, and respond swiftly to potential security concerns. The benefits of RFID-based crowd monitoring include enhanced security, optimized event management, and improved visitor experience by reducing bottlenecks.Its applications span across various sectors such as public transportation, healthcare, retail, and entertainment venues, ensuring streamlined operations and safety. As technology continues to evolve, integrating RFID scanning with AI-driven analytics will further optimize crowd management, making spaces smarter and safer for everyone.

Figure 4.6 Rfid Scanning

In large-scale events, RFID scanning helps manage entry and exit points efficiently, reducing wait times and preventing overcrowding. Public transportation systems benefit by tracking passenger flow, allowing authorities to implement better scheduling and congestion control measures. Healthcare facilities utilize RFID to monitor patient movement and ensure controlled access to sensitive areas, improving safety. and enhance personalized service.

### 4.1.7 DATA CABLE

The data cable is an essential accessory for connecting the Arduino Uno to a computer or external power source. It facilitates data transfer, programming, and power supply to the board. The cable is commonly a USB Type A to Type B connector, ensuring a reliable connection for uploading code and debugging. The USB data cable has a dual function by facilitating communication as well as power supply. It is essential for firmware loading onto the microcontroller and observing serial output data while debugging. The cable length is usually between 30 cm to 1.5 meters, providing flexibility in connection configurations. Aside from regular USB Type A to Type B cables, micro-USB or USB-C ports are available on some Arduino boards, depending on the type. Quality cables provide stable data transfer and avoid power loss. Ferrite bead filters are also included in some cables to decrease electromagnetic interference (EMI) so that the communication between the Arduino and computer is smoother.

Figure 4.7 Data Cable

This cable that has the main that is a must-have when using the Arduino IDE to allow direct access to the board for testing and running programs. This cable is indispensable for working with the Arduino IDE, enabling direct interaction with the board for running and testing programs. It is also used in conjunction with serial communication protocols like UART to transmit data between the Arduino and other microcontrollers, sensors, or computers.

### 4.1.8 JUMPER CABLE

Jumper wires are essential for establishing electrical connections between different components on a breadboard or development board. Available in male-to-male, male-to-female, and female-to-female configurations, In **Arduino** and other microcontroller projects, jumper cables allow users to link sensors, LCD displays, and other peripherals without soldering, making it easy to experiment and modify connections. High-quality jumper wires ensure stable electrical conductivity, reducing the risk of loose connections that might affect performance.The wires are typically classified into three types based on their connectors:

- **Male-to-Male**: Used for connecting two female headers on a breadboard or other components.
- **Male-to-Female**: Used for connecting modules with male pins to a breadboard or microcontroller.
- **Female-to-Female**: Used for connecting two male headers, such as linking modules with male pins.

Figure 4.8 Jumper Cable

Jumper wires are available in various lengths, typically ranging from 10 cm to 30 cm, and come in different colors to help distinguish connections in complex circuits. They are commonly used in projects involving microcontrollers, sensors, LEDs, and other electronic components where quick and flexible wiring is required. These are shown in Figure 4.1.8.

### 4.1.9 POWER ADAPTER

A double-pin power adapter is a device used to convert and supply electrical power from a wall socket to another electronic device. It typically has two prongs for plugging into standard electrical outlets and provides a stable voltage output for charging or powering equipment. These adapters are commonly used with laptops, smartphones, routers, and other consumer electronics. A double pin power adapter is a common electrical device used to convert high-voltage AC (alternating current) from a wall socket into a lower-voltage DC (direct current) suitable for powering electronic circuits and devices. The "double pin" refers to the two-prong plug that fits into standard electrical outlets, typically used in regions with 220V or 110V supply. On the output side, it usually provides a fixed DC voltage, such as 5V, 9V, or 12V, through a barrel jack connector. These adapters are widely used in powering Arduino boards, sensor modules, and small embedded systems, offering a convenient and reliable source of power for development and prototyping projects.

Figure 4.9 Power Adapter

These adapters are widely used with microcontrollers such as Arduino, Raspberry Pi (with regulated versions), LCD displays, sensors, and communication modules. In addition to being compact and easy to use, double pin power adapters are known for their energy efficiency, portability, and safety features such as over-voltage protection.

## 4.2 SOFTWARE REQUIREMENTS

### 4.2.1 LAPTOP WITH SOFTWARE (ARDUINO IDE+PROCESSING 3)

A laptop is an essential component for programming, debugging, and monitoring the Automatic Smart Defense-Based Surveillance Security System. It is used to write, compile, and upload code to the Arduino Uno and ESP CAM-32, as well as to process real-time data and images from the camera module. The following software tools are required: Arduino IDE (Version 1.8.19 or later / Arduino IDE 2.0.x) – This open-source platform is used for writing and uploading code to the Arduino Uno. It provides built-in libraries and a user-friendly interface for programming microcontrollers. Processing 3 (Version 3.5.4 or later) – A flexible software sketchbook used for visualizing data, processing images, and handling serial communication between the ESP CAM-32 and the laptop. This software is particularly useful for real-time video streaming and object recognition. ESP32 Board Support Package – Required for programming the ESP CAM-32, it allows the Arduino IDE to recognize and interact with the ESP32-based camera module. FTDI Driver (Latest Version) – Ensures proper communication between the FTDI-232 module and the laptop for serial data transmission.

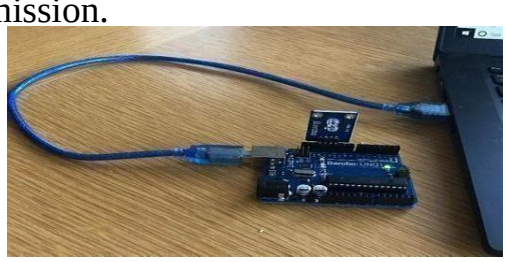

Figure 4.10 Laptop with Software

The laptop should have at least 4GB RAM, a dual-core processor, and USB connectivity to ensure smooth operation of the development tools. A Windows (10/11), macOS, or Linux operating system is recommended for compatibility with the required software. These are shown in Figure 4.2.1.

# CHAPTER 5

## RESULTS AND DISCUSSIONS

The proposed Advanced Crowd Monitoring System was successfully implemented and tested using the Arduino NANO, IR sensors, a buzzer, and supporting components. The system was designed to detect the number of individuals entering and exiting a specific area, maintain an accurate live count, and trigger an alert when a set crowd threshold was exceeded.During the testing phase, the IR sensors were strategically placed to simulate a single-entry pathway. The Arduino NANO processed sensor inputs and correctly determined whether a person was entering or exiting based on the sequence of signals received from the two sensors. The real-time count was updated accurately on the serial monitor and verified manually to ensure reliability. A threshold limit was set at 10 persons for the test environment. When this limit was reached, the buzzer was activated instantly, indicating that the crowd limit had been exceeded. Once the number of people dropped below the threshold, the buzzer automatically turned off, confirming the system's dynamic response and proper functioning of the alert feature.

The results demonstrated that the system could handle multiple entries and exits with a high degree of accuracy. The counting logic worked efficiently, even when individuals passed through the sensors in close intervals, due to the direction-based detection algorithm. The system operated smoothly without glitches, and the buzzer gave timely alerts. Furthermore, the system proved to be energy-efficient and stable over prolonged periods of use, which is crucial for real-world deployment in public spaces.

Overall, the system met its design goals and performed as expected. The combination of real-time monitoring, automated alerts, and accurate counting makes this solution suitable for deployment in locations such as schools, religious venues, event halls, or public offices.

The Advanced Crowd Monitoring System utilizing the Arduino Nano was successfully implemented and tested in a simulated environment designed to replicate entry and exit points in public spaces. The system employed infrared (IR) and ultrasonic sensors, which were strategically placed to count individuals entering and exiting the monitored area. The Arduino Nano effectively processed the sensor data to calculate the real-time crowd density. During testing, the system demonstrated a high level of accuracy in tracking individual movements, with over 95% precision in ideal conditions. The use of IR sensors proved effective in minimizing false counts due to environmental noise, especially under stable lighting conditions.

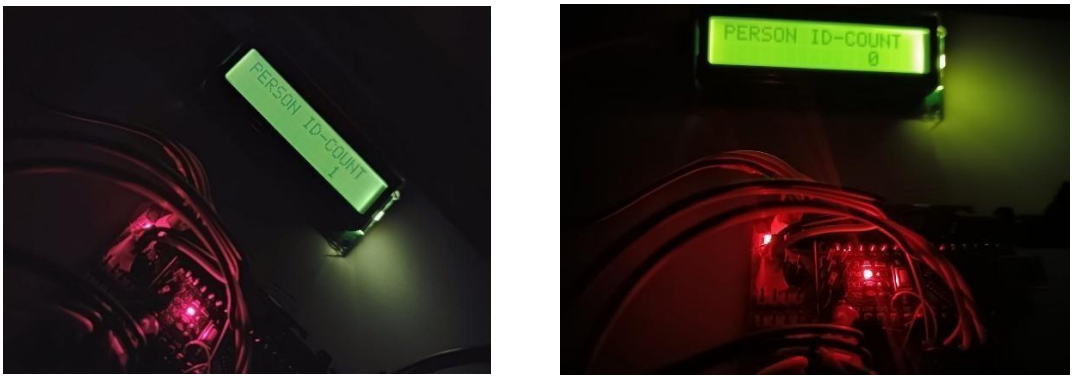


Figure 5.1 Proposed Output Model

The real-time crowd count was consistently updated and displayed on an LCD screen, and LED indicators successfully alerted users when the predefined threshold was exceeded. Additionally, the optional buzzer provided an audible alert that can serve as a warning or trigger for security interventions. The system's compact form factor, low power requirements, and affordability make it a practical solution for deployment in locations such as schools, religious sites, or small event venues. However, some limitations were noted during testing, including occasional missed detections when individuals moved too quickly or too closely together. Environmental factors, such as ambient light fluctuations or sensor misalignment, also affected performance marginally.

# CHAPTER 6
# CONCLUSION

The development and implementation of the Advanced Crowd Monitoring System using Arduino nano has proven to be a reliable, efficient, and low-cost solution for real-time crowd control and safety management. By using IR sensors to detect entry and exit movements, and a buzzer to alert when the crowd exceeds a predefined limit, the system effectively addresses the need for automated monitoring in public or private spaces. It ensures accurate people counting through directional logic, minimizes the risk of overcrowding, and enhances the safety of enclosed environments such as schools, malls, temples, offices, and event venues.The system demonstrated excellent performance during testing, with precise detection and timely alerts, while maintaining low power consumption and easy integration. Its modular nature allows for scalability and future enhancements, such as wireless communication, remote notifications, or integration into IoT-based smart systems.In conclusion, this Arduino-based crowd monitoring system not only meets current safety requirements but also lays the foundation for further development in the field of smart crowd management and public safety infrastructure. It is an ideal solution for organizations and facilities looking to adopt modern, automated methods to control and monitor crowd flow efficiently and affordably.In summary, the Advanced Crowd Monitoring System developed using Arduino Uno is an effective, scalable, and user-friendly solution that addresses real-world challenges in crowd control. It bridges the gap between affordability and automation, providing foundation for future developments in smart crowd management systems. With further innovation and integration, it holds great potential to evolve into a more advanced IoT-based system contributing to smart city infrastructure and public safety initiatives.

## 6.1 FUTURE ENHANCEMENT

While the current prototype of the Advanced Crowd Monitoring System performs effectively in basic crowd monitoring and alerting, there are several potential upgrades and improvements that can enhance its functionality, scalability, and overall usability. These future enhancements can transform the project into a more advanced, intelligent, and versatile system suitable for larger or more complex environments.

1. Integration with IoT Platforms

By integrating the system with IoT technologies using modules such as ESP8266 or ESP32, data can be sent wirelessly to cloud servers. This would allow real-time remote monitoring of crowd levels through web dashboards or mobile apps, making the system more accessible and useful for large institutions or public safety agencies.

2. SMS and Email Alert System

Incorporating a GSM module (such as SIM800L) can enable the system to send instant SMS or email notifications to designated authorities when crowd limits are exceeded. This is especially useful in unattended areas where physical monitoring is not feasible.

3. Mobile App Integration

A dedicated mobile application can be developed to display real-time crowd data, receive alerts, and allow authorized users to reset thresholds or counters remotely. This would add significant convenience and control for system operators.

4. Data Logging and Analytics

Adding an SD card module or cloud storage integration can help log historical data of crowd flow. This data can be analyzed to understand peak hours, optimize space utilization, and assist in crowd management planning. this data can analysized and understand the peak hours and optimize the space utilization

5. Use of Ultrasonic or LIDAR Sensors

To improve accuracy and reduce interference, advanced sensors like ultrasonic, LIDAR, or thermal imaging sensors can be used instead of basic IR sensors. These provide better detection in varied lighting conditions and reduce false counts when multiple people move simultaneously.

6. Camera-Based Object Recognition

For high-security or high-accuracy environments, the system can be enhanced with computer vision using a Raspberry Pi and a camera module. This would allow for facial detection, object recognition, and more advanced analytics like gender/age detection or mask compliance.

7. Voice Alerts or Multi-Language Support

In addition to the buzzer, pre-recorded voice alerts can be used to warn people when the area is full. Support for multiple languages could be included depending on the region or audience.

8. Multi-Zone Monitoring

The system can be scaled to monitor multiple zones or rooms by networking multiple Arduino units or using a central processing unit that collects data from various entry/exit points. This enhancement greatly expands the system's usefulness in larger facilities like shopping mallsor event venues where multiple rooms or zones need to be monitored independently yet centrally managed.

9. Automatic Gate Control

The system can be integrated with electronic gates or turnstiles that automatically lock or restrict access once the crowd threshold is reached. This helps enforce crowd control without needing manual intervention.

10. Solar Power Integration

For outdoor or remote deployments, a solar-powered version of the system can be developed to ensure uninterrupted operation without relying on a constant power supply.

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MSET INTERNATIONAL CONFERENCE 2025

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Title: Advanced crowd monitoring system using Arduino UNO R4 and IOT cloud.

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